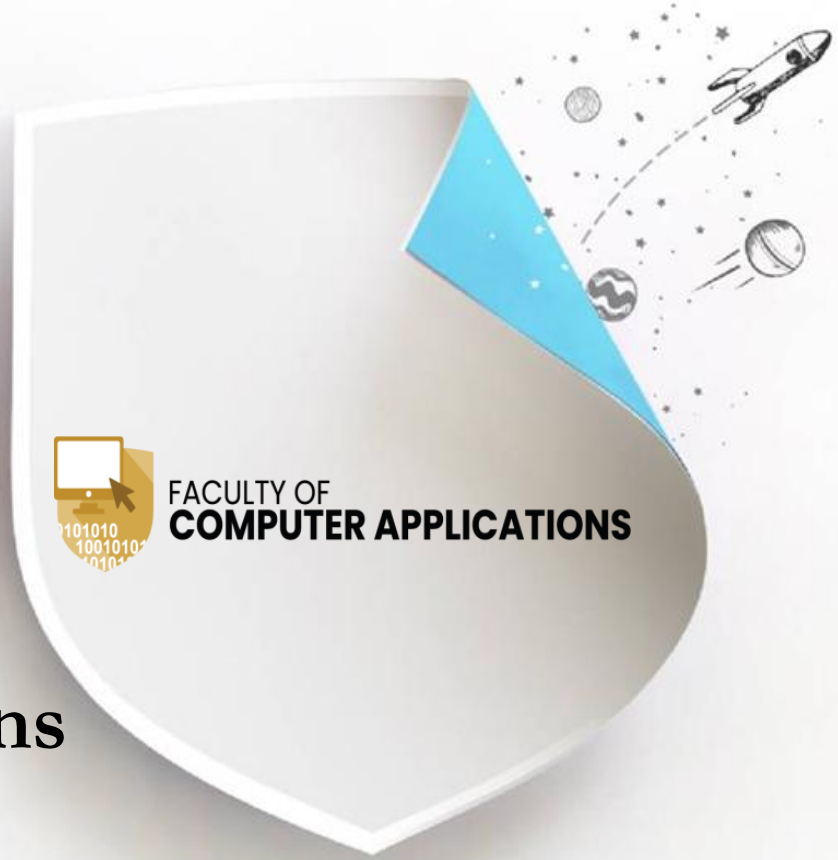
Marwadi University
Marwadi Chandarana Group

NAAC A+



FACULTY OF COMPUTER APPLICATIONS

Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems

PROGRAM	BACHELOR OF COMPUTER APPLICATIONS
SEMESTER	2
COURSE TITLE	ANALYSIS AND DESIGN OF SYSTEMS
COURSE CODE	05BC3202
COURSE CREDITS	4
COURSE DURATION	20 HOURS

UNIT 4 – DESIGN OF SYSTEM

TOPICS COVERED

- Designing Output
  - Output Design Objectives
  - Relating Output
  - Content To Output Method
  - Realizing Output Bias Effects
  - Designing Output For Displays
  - Designing Output For Websites
- Designing Input
  - Good Form Design,
  - Good Display And Web Forms Design,
  - Intranet And Internet Page Design

TOPICS COVERED

- Designing Database
  - Databases,
  - Data Concepts,
  - Normalization,
  - Guidelines For Master File/Database Relation Design,
  - Denormalization,
  - Data Warehouse
- Human Computer Interaction(HCI)
  - Understanding HCI,
  - Usability,
  - Types Of User Interface,
  - Guidelines For Dialog Design,
  - Feedback For Users

Designing Output

- The design of output is the most important task of any system. During output design, developers **identify the type of output needed**.

- **Output is information delivered to users through the information system by way of intranets, extranets, or the World Wide Web computer.**

- Output is the most important & direct source of information for the user. The system is accepted by the user only by the quality of its output. If the output is not of good quality, the user is likely to reject the system. Therefore, **an effective output design is a major criterion for deciding the system's quality**.

Designing Output

While designing the output one should try to accomplish the following:

- Determine **what information to present** to the user.
- Decide **whether to display or print** the information.
- **Arrange the presentation of information** in an acceptable format etc

OUTPUT DESIGN OBJECTIVES

- There are six objectives that the systems analyst tries to attain when designing output:

1. Designing output to serve the intended purpose.
2. Designing output to fit the user.
3. Delivering the appropriate quantity of output.
4. Making sure the output is where it is needed.
5. Providing the output on time.
6. Choosing the right output method.

OUTPUT DESIGN OBJECTIVES

1. **Designing output to serve the intended purpose :**

All output should have a purpose. During the information requirements determination phase of analysis, the systems analyst finds out what user and organizational purposes exist. Output is then designed based on those purposes. Remember the **rule of purposiveness,** however. If the output is not functional, it should not be created, because there are costs of time and materials associated with all output from the system.

OUTPUT DESIGN OBJECTIVES

2. **Designing output to fit the user:**

With a large information system serving many users for many different purposes, it is often difficult to personalize the output. On the basis of interviews, observations, and cost considerations, it will be possible to design an output that addresses what users need and prefer. Generally speaking, it is more practical to create **user-specific or user-customizable output** when designing for a decision support system or other highly interactive.

OUTPUT DESIGN OBJECTIVES

3. **Delivering the appropriate quantity of output :**

Part of the task of designing output is deciding what quantity of output is correct for users. A useful heuristic is that the system must provide what each person needs to complete his or her work means information at first and then provide a way for the user to access additional information easily. Often they will not need great amounts of output, especially if there is an easy way to access more via a hyperlink or drill-down capability.

OUTPUT DESIGN OBJECTIVES

4. **Make sure the output is where it is needed:**

Output is often produced at one location and then distributed to the user. Output must be presented to the right user at right time.

OUTPUT DESIGN OBJECTIVES

5. **Providing the output on time :**

One of the most common complaints of users is that they do not receive information in time to make necessary decisions. Many reports are required on a daily basis, some only monthly, others annually, etc. Accurate timing of output can be critical to business operations.

OUTPUT DESIGN OBJECTIVES

6. **Choosing the right output method :**

Choosing the right output method for each user is another objective in designing output. Much output now appears on display screens, and users have the option of printing it out with their own printer. The analyst needs to recognize the trade-offs involved in choosing an output method. Costs differ for the user, there are also differences in the accessibility, flexibility, durability, distribution, storage, retrieval possibilities, transportability, and overall impact of the data.

REALISING OUTPUT BIAS EFFECTS AND AVOID BIASING

- Bias is present in everything that humans create. This statement is not to judge bias as bad, but to make the point that it is inseparable from what we (and consequently our systems) produce.
- The concerns of systems analysts are to avoid unnecessarily biasing output and to make users aware of the possible biases in the output they receive.

Presentations of output are unintentionally biased in three main ways:

1) How information is sorted.
2) Setting of acceptable limits.
3) Choice of graphics.

REALISING OUTPUT BIAS EFFECTS AND AVOID BIASING

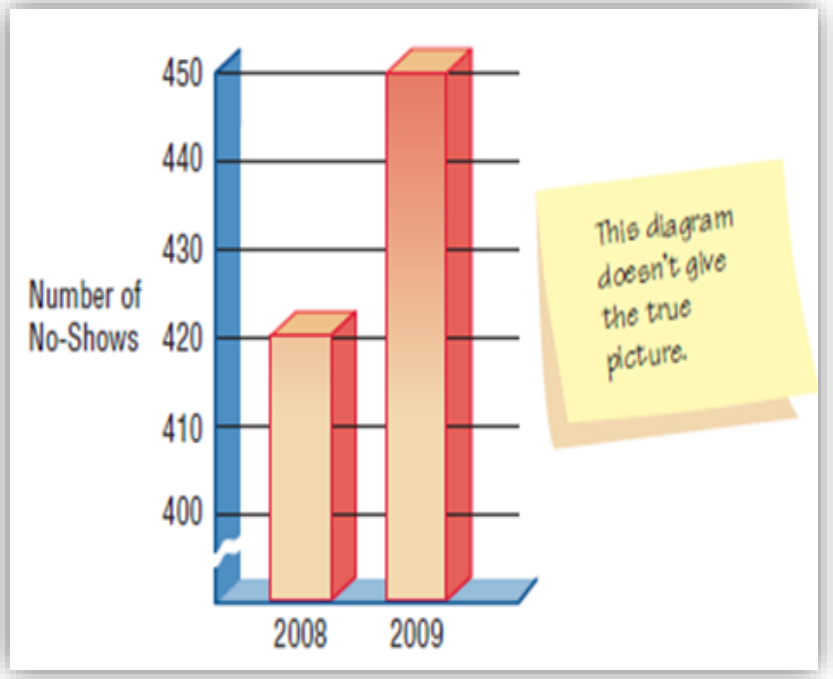
1. **How information is sorted :** Bias is introduced to output when the analyst and users make choices about how information is sorted for a report. Common sorts include alphabetical, chronological, and cost. For example, if past suppliers are listed alphabetically, companies such as Aardvark Printers, Advent Supplies, and Barkley Office Equipment are shown to the purchasing manager first.

2. **Setting of acceptable limits:** Second major source of bias in output is the predefinition of limits for particular values being reported. Many reports are generated on an exception basis only, which means, only exceptions to those values will be output.

REALISING OUTPUT BIAS EFFECTS AND AVOID BIASING

3. **Choices of graphics :**

Bias can occur in the selection of the graph size, its color, the scale used, and even the type of graphic. Graph size must be proportional so that the user is not biased as to the importance of the variables that are presented. For example, the Figure shows a column chart comparing the number of no-shows for hotel bookings in 2008 with no-shows in 2009. Notice that the vertical axis is broken, and it appears that the number of no-shows for 2008 is twice as much as the number of no-shows in 2009, although the number of no-shows has gone up only slightly.

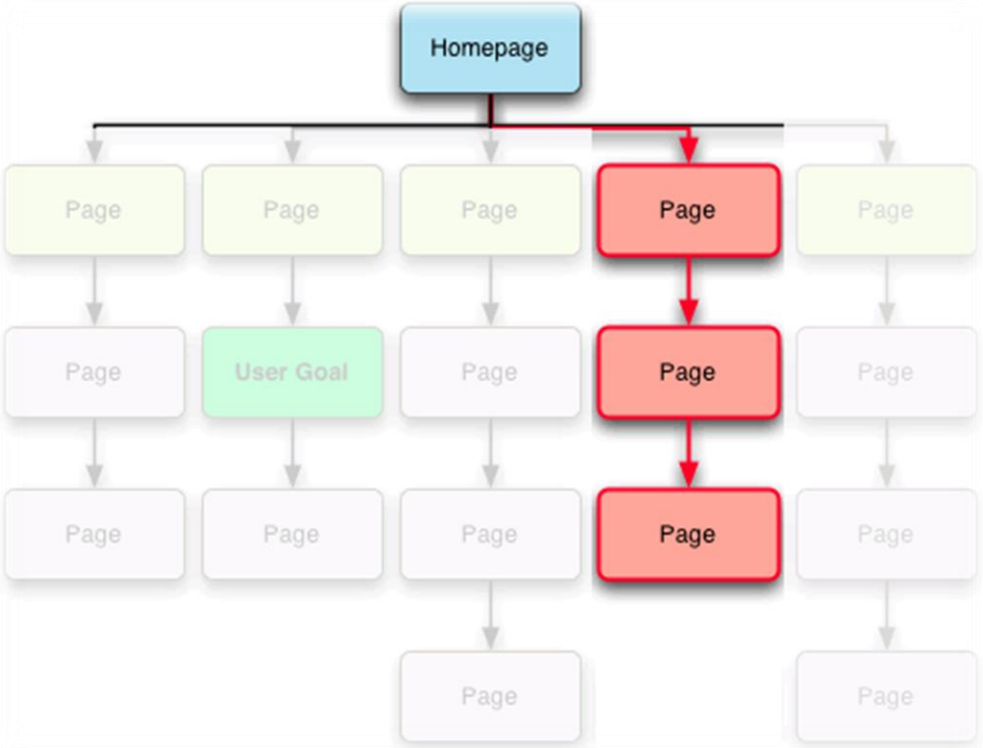


DESIGNING OUTPUT FOR DISPLAYS AND WEBSITE

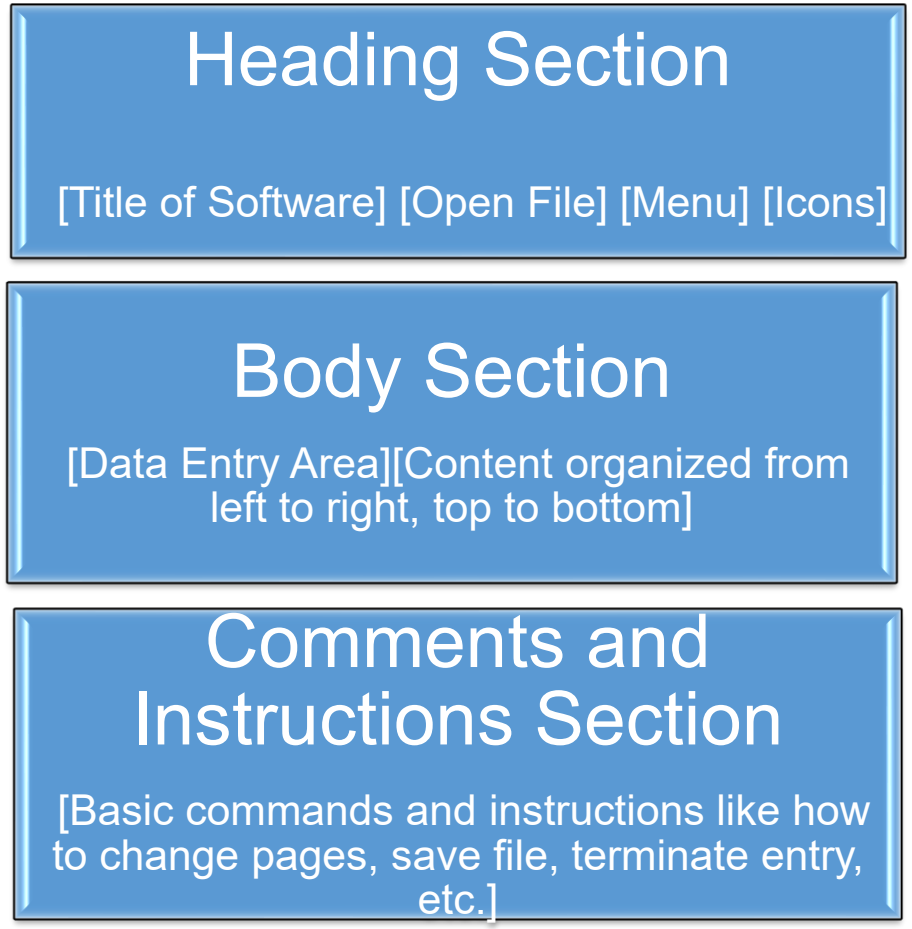
Guidelines for Display Design

Four guidelines facilitate the design of displays:

1) Keep the display simple.
2) Keep the presentation consistent.
3) Facilitate user movement among displayed output.
4) Create an attractive and pleasing display.



GUIDELINES FOR DISPLAY DESIGN



1. **Keep the display simple :**

Display output should be divided into three sections. The top of the screen features a "heading" section.

- The **heading** contains titles of software and open files, pulldown menus, and icons that do certain tasks.

- The **middle** section is called the "body" of the display. The body can be used for data entry and organized from left to right and top to bottom because people in Western cultures move their eyes on a page in this way.

GUIDELINES FOR DISPLAY DESIGN

2. **Keep the display presentation consistent:**

- Displays can be kept consistent by locating information in the same area each time a new display is accessed.

- Also, information that logically belongs together should be consistently grouped together:
  Ex. Name and address go together, not name and zip code.

- Although the display should have a natural movement from one region to another, information should not overlap from one group to another. You would not want a name and address in one area and a zip code in another.

GUIDELINES FOR DISPLAY DESIGN

3. **Facilitate user movement among display screens and pages:**

- The third guideline for good display design is to make it easy to move from one page to another. The **"three-clicks" rule** says that users should be able to get to the pages they need within three mouse or keyboard clicks.



GUIDELINES FOR DISPLAY DESIGN

4. **Create an attractive and pleasing display:**

- The fourth guideline for good display design is to create an attractive display for the user. Use logical flows in the plan for your display pages. With the advent of GUIs, it is possible to make input displays very attractive. By using color or shaded boxes and creating three-dimensional boxes and arrows, you can make forms user-friendly and fun to use. When contemplating the use of different font styles and sizes, ask yourself if they truly assist the user in understanding and approving the display.

- If they draw undue attention to the art of display design or if they serve as a distraction, leave them out. Be aware that not all Web pages are viewed identically by different browsers. Test your prototype forms with a variety of combinations to see if users declare preferences for combinations or whether they are distressing to the majority of users. For Web fonts use Verdana or Arial.

DESIGNING OUTPUT FOR DISPLAYS, WEBSITES

Guidelines for WEBSITE DESIGNING

1. Use professional tools
2. Study other websites
3. Use the resources that the web has to offer
4. Examine the websites of professional designers
5. Use the tools you've learned
6. Consult the books
7. Look at some poor examples of web pages, too.
8. Create templates of your own
9. Use plug-ins, audio, and video very sparingly
10. Plan ahead

DESIGNING INPUT

- Users should receive high-quality results. The caliber of a system's output is determined by its input quality.
- Efficient input forms, displays, and interactive Web fill-in forms should be created with consistency, accuracy, ease of use, simplicity, and aesthetics in mind.
- All of these goals may be achieved by applying fundamental design principles, being aware of the input requirements for the system, and comprehending how users react to various form and display elements.

WELL DESIGNED INPUT OBJECTIVES

1. Effectiveness
2. Accuracy
3. Ease of use
4. Consistency
5. Simplicity
6. Attractiveness
7. Control the amount of input: Minimize the quantity of data for input
8. and avoid duplication in data collection and data entry.
9. Prevent data errors by limiting the amount of input,
10. Keep the process simple

GOOD FORMS DESIGN GUIDELINES

- The systems analyst should be capable of designing a complete and useful form. To design forms that people find useful, four guidelines for form design should be observed:

1. **Make forms easy to fill in.**
2. **Ensure that forms meet the purpose for which they are designed.**
3. **Design forms to ensure accurate completion.**
4. **Keep forms attractive.**

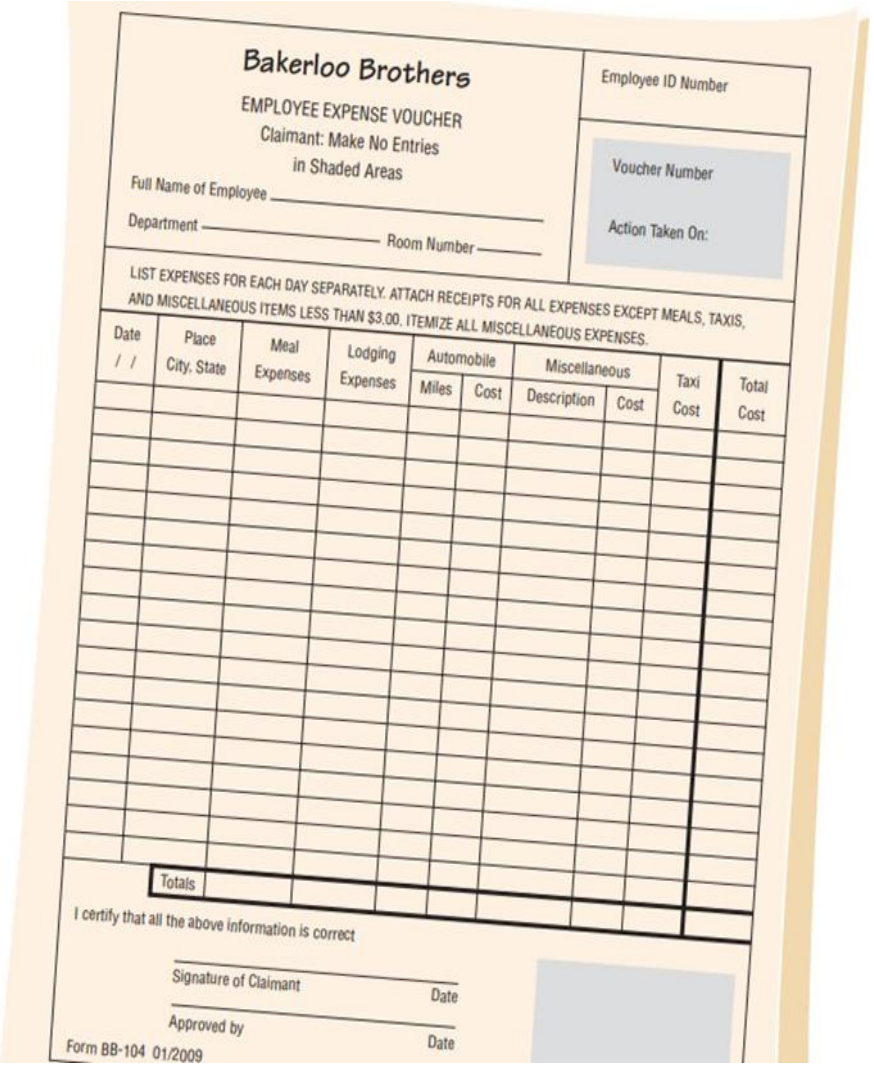
GOOD FORMS DESIGN GUIDELINES

1. **MAKE FORMS EASY TO FILL IN:** To reduce error, speed completion, and facilitate the entry of data, forms must be easy to fill in.

A. **FORM FLOW:** Illogical flow takes extra time and is frustrating. Forms should flow from left to right and top to bottom.

B. **SEVEN SECTIONS OF A FORM:** A second method that makes it easy for people to fill out forms correctly is the logical grouping of information. The seven main sections of a form are the following: 1. Heading. 2. Identification and access. 3. Instructions. 4. Body. 5. Signature and verification. 6. Totals. 7. Comments. Notice that the seven sections cover the basic information required on most forms. (FIGURE IS GIVEN IN NEXT SLIDE)

C. **CAPTIONING:** Clear captioning is another technique that can make easy work of filling out a form. Captions tell the person completing the form what to put in a blank line, space, or box



Bakerloo Brothers

EMPLOYEE EXPENSE VOUCHER
Claimant: Make No Entries in Shaded Areas

Employee ID Number

Voucher Number

Action Taken On:

Full Name of Employee

Department

Room Number

LIST EXPENSES FOR EACH DAY SEPARATELY. ATTACH RECEIPTS FOR ALL EXPENSES EXCEPT MEALS, TAXIS, AND MISCELLANEOUS ITEMS LESS THAN $3.00. ITEMIZE ALL MISCELLANEOUS EXPENSES.

Date / /	Place City, State	Meal Expenses	Lodging Expenses	Automobile		Miscellaneous		Taxi Cost	Total Cost
				Miles	Cost	Description	Cost		
Totals									

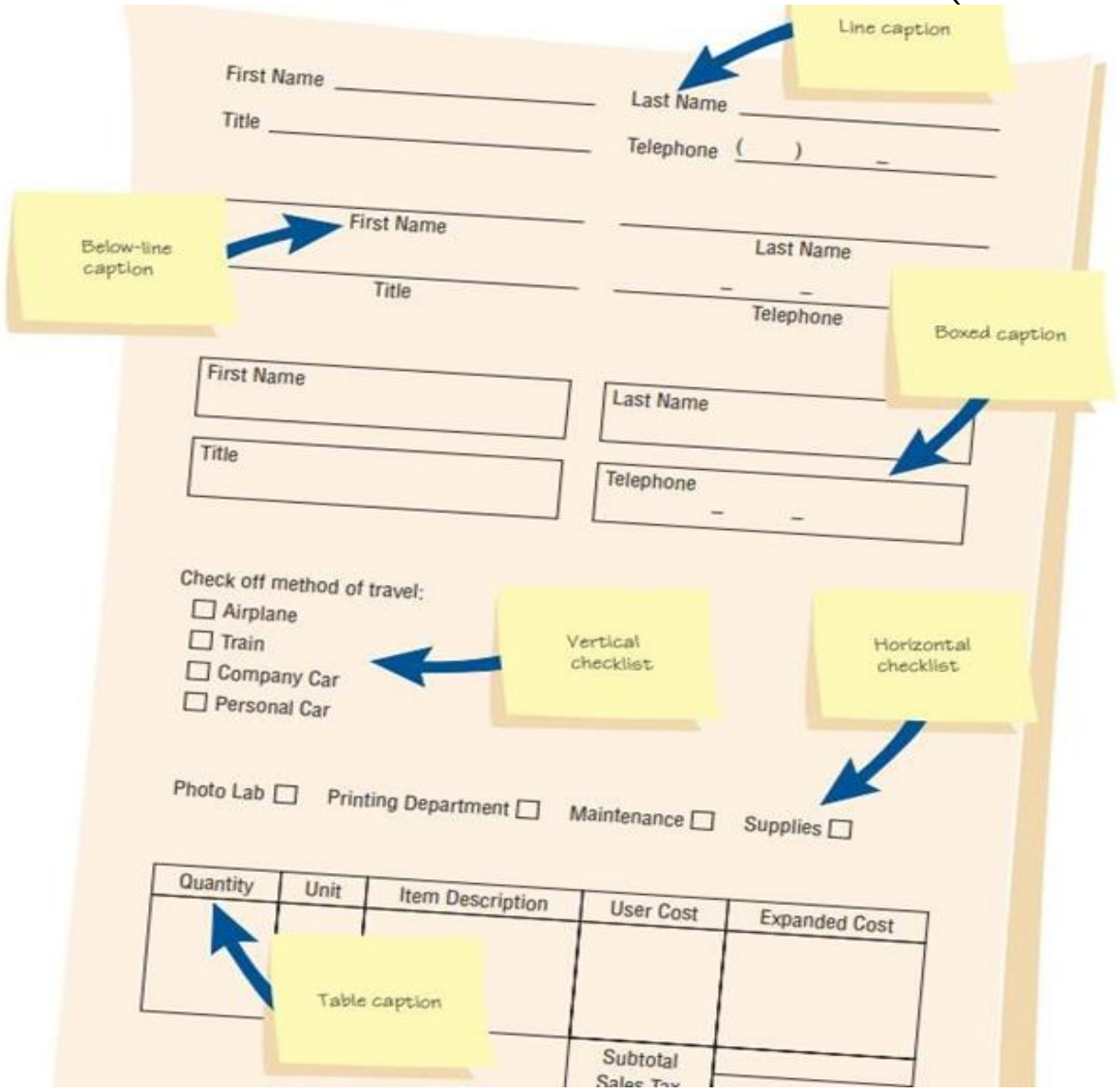
I certify that all the above information is correct

Signature of Claimant Date

Approved by Date

Form BB-104 01/2009

GOOD FORMS DESIGN GUIDELINES(CAPTIONS IN FORM)



GOOD FORMS DESIGN GUIDELINES

2. **Meeting The Intended Purpose:**

Ensure that forms meet the purpose for which they are designed.

3. **Ensuring Accurate Completion:**

Design is important for ensuring that people do the right thing with the form whenever they use it.

GOOD FORMS DESIGN GUIDELINES

4. **Keeping Forms Attractive:**

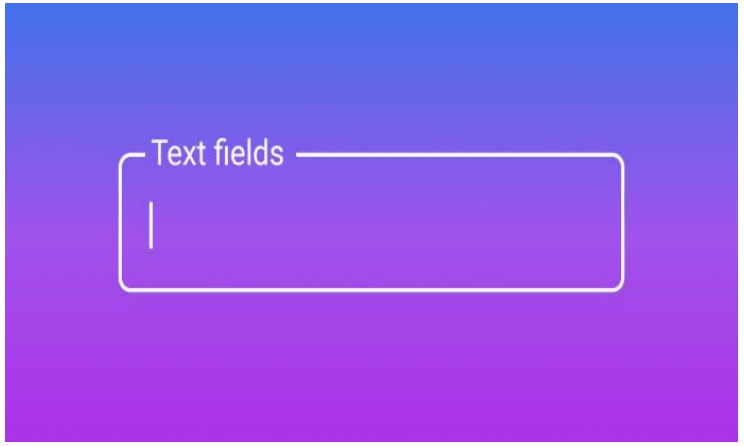
Using different types of fonts in the same form can help make it appealing for users to fill in. Separating categories and subcategories with thick and thin lines can also encourage interest in the form. Type fonts and line weights are useful to design elements for capturing attention and making people feel secure that they are filling in the form correctly.

5. Controlling Business Forms: The basic duties for controlling forms include making sure that each form in use fulfills its specific purpose in helping workers accomplish their tasks at the lowest cost.

GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)

❑ Graphical User Interface Design:

A graphical user interface is a way that users interface with the OS. This is also referred to as a point-and-click interface that takes advantage of additional features in display design such as text boxes, check boxes, option buttons, list and drop-down list boxes, sliders and spin buttons, tab control dialog boxes, and image maps.

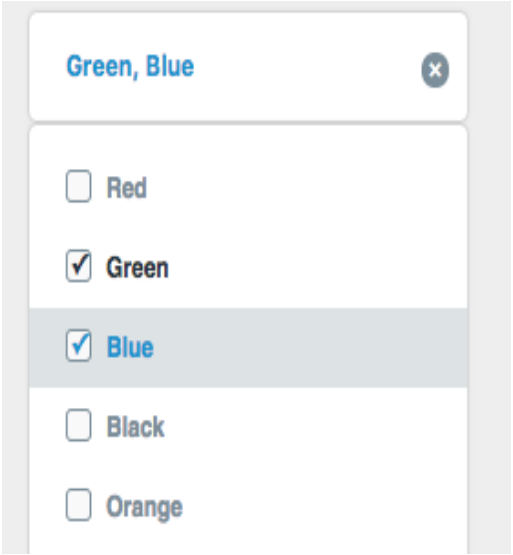


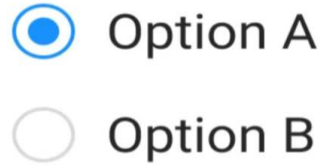
A. **Text Boxes**: It is used to outline data entry and display fields. Care must be taken to ensure that the text box is large enough to accommodate all the characters that must be entered. Each text box should have a caption to the left, identifying what is to be entered or what is displayed in the box.

GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)

B. Check Boxes:

They are used for nonexclusive choices in which one or more of the options may be checked. If there is more than one check box, the labels should have some order to them, either alphabetic or with the most commonly checked item appearing first in a list. If there are more than 10 checkboxes, group them in a bordered box.





C. Option Buttons:

A circle called an option button or a radio button is used to select exclusive choices. Only one of several options can be chosen. If there is a commonly selected option, it is usually selected as a default when the page first displays. If there are more than six option buttons, consider using a list box or a drop-down list box.

GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)

D. List And Drop-down List Boxes:

A list box displays several options that may be selected with the mouse. A single rectangle with an arrow points down toward a line located on the right side of the rectangle. Selecting this arrow causes a list box to be displayed. Once a user makes a choice, it is displayed in the drop-down selection rectangle.

E. Tab Control Dialog Boxes:

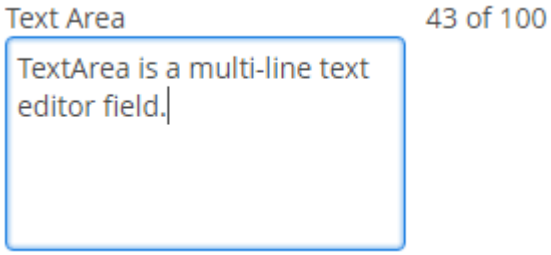
In designing tab control boxes create a separate tab for each unique feature, place the most commonly used tabs in front and display them first, and include buttons for OK, Cancel, and Help

GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)



F. Image Maps :

Image maps are used when creating Web pages containing maps with instructions to click in a certain area in order to view a detailed map of the region.



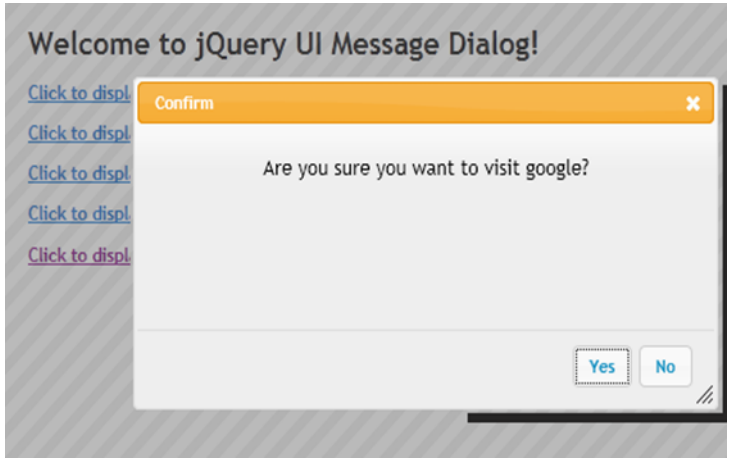
G. Text Areas:

A text area is used for entering a larger amount of text. These areas include several rows, columns, and scroll bars that allow the user to enter and view text greater than the size of the box area. There are two ways to handle this text. One is to avoid the use of word wrap, forcing the user to press the Enter key to move to the next line; the text will scroll to the right if it exceeds the width of the text area.

GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)

H. Message Boxes:

Message boxes are used to warn users and provide other feedback messages in a dialog box, often overlapping the display. These message boxes have different formats. Each should appear in a rectangular window and should clearly spell out the message so that the user knows precisely what is happening and what actions are possible.



GOOD DISPLAY AND WEB FORMS DESIGN,(GUI)

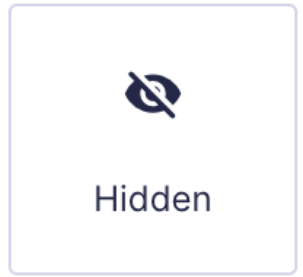
I. Command Buttons :

A command button performs an action when the user selects it with the mouse. Calculate **Total, Add Order, and OK** are all examples. The text is centered inside the button which has a rectangular shape.



J. Hidden Fields:

These are not visible to the viewer, do not take up any space on the Web page, and can contain only a name and a value. Often hidden fields are used to store values sent from one Web form to the server. Sometimes a hidden field will contain a key field used to locate a record for the customer or the browsing session.

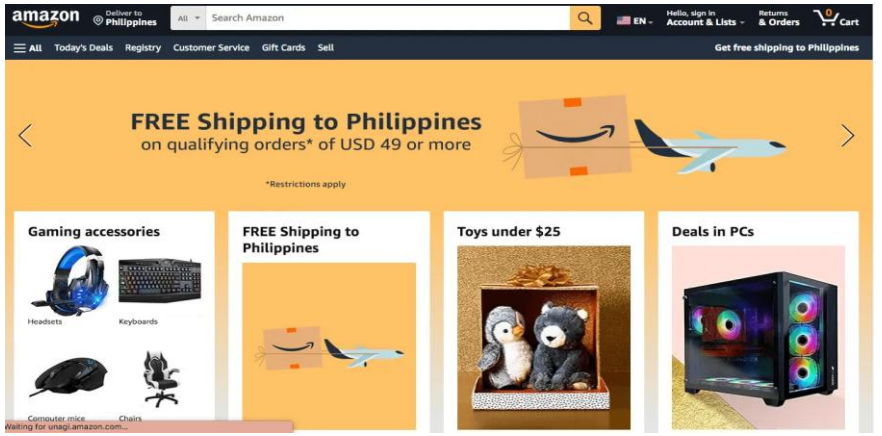


GOOD DISPLAY AND WEB FORMS DESIGN

Dynamic Web Pages :

- Dynamic Web pages change themselves as the result of user action.

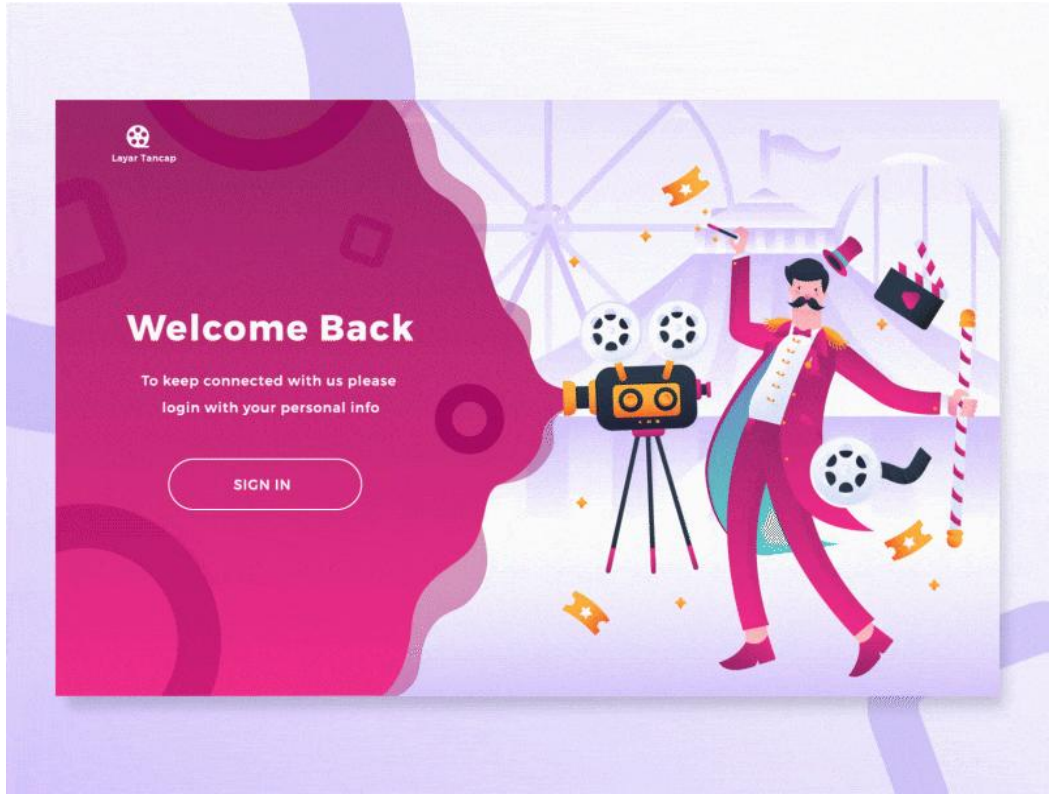
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GOOD DISPLAY AND WEB FORMS DESIGN

Dynamic Web Pages :

- They often use JavaScript to modify some part of the Web page or a style. Changing an image when the mouse moves over it or rotating random images at a given time interval are common examples of dynamic Web pages.

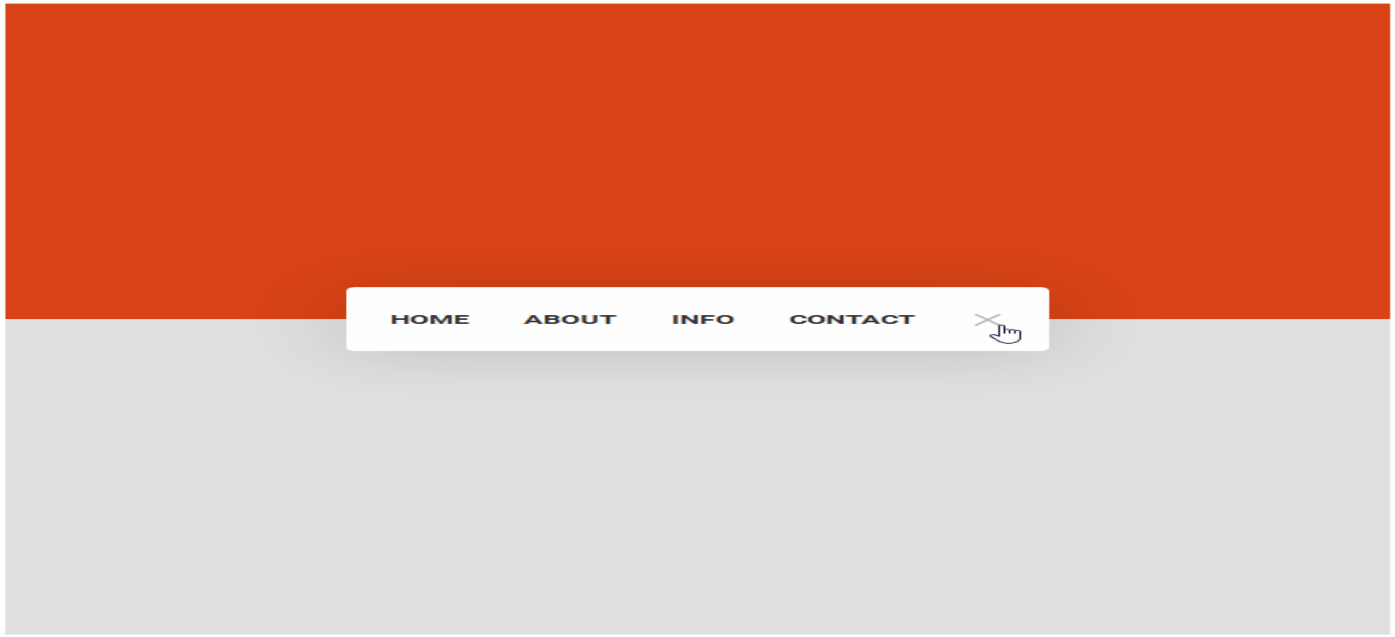


GOOD DISPLAY AND WEB FORMS DESIGN

Dynamic Web Pages :

- Menus that expand when the user clicks a small plus sign to the left of the menu or when the mouse moves over a menu are other examples of dynamic Web pages.





INTERNET AND INTRANET PAGE DESIGN

- **What is Intranet?**

An intranet is a private network that gives employees in a company the ability to organize information, manage documents, and shares calendars and to-do list. It normally runs in a client/server environment in a local area network. In Intranet, every computer is connected via the LAN and has something known as a MAC address. It is a number that allows you to identify the place where the computer is located.



INTERNET AND INTRANET PAGE DESIGN

- **What is the Internet?**

The internet is a collection of interconnected devices that are spread across the globe. The internet is a network of networks that consist of public, private, sales, finance, academic, business, and government networks. The internet is a type of network called a network of networks.

INTERNET AND INTRANET PAGE DESIGN

- Here are more hints about designing a good Internet or intranet fill-in form that should be noted now that you have learned some of the elementary aspects of input form and display design. **Guidelines include the following:**

1. Provide clear instructions, because Web users may not be familiar with technical terminology.

2. Demonstrate a logical entry sequence for fill-in forms, especially because the users may have to scroll down to a region of the page that is not visible at first.

3. Use a variety of text boxes, push buttons, drop-down menus, checkboxes, and radio buttons to serve specific functions and to create interest in the form.

INTERNET AND INTRANET PAGE DESIGN

4. Provide a scrolling text box if you are uncertain about how much space users will need to respond to a question, or about what language, structure, or form users will use to enter data.

5. Prepare two basic buttons on every Web fill-in form Submit and Clear Form.

6. If the form is lengthy and the users must scroll excessively, divide the form into several simpler forms on separate pages.

7. Create a feedback screen that refuses submission of a form unless mandatory fields are filled in correctly. The returned form screen can provide detailed feedback to the user in a different color. Red is appropriate here.

DESIGNING DATABASE

- Data storage is considered by some to be the heart of an information system that allows the creation, modification, and updating of the database; the retrieval of data; and the generation of reports and displays it.

- The person who ensures that the database meets its objectives I called the database administrator.

Reality, Data, and Metadata :

- The real world will be referred to as reality
- Collection of records is called data.
- Data about data is called metadata. It describes relevant information about data.

DESIGNING DATABASE

Rollno	Name	Result
1	A	PASS
2	B	FAIL

METADATA

DATA

ENTITIES

- The entity in DBMS can be a real-world object with an existence. An entity may be a person, place, or thing. For example, in a College database, the entities can be Professors, Students, Courses, etc.

- An entity subtype is a special one-to-one relationship used to represent additional attributes (fields) of another entity that may not be present on every record of the first entity.

ENTITIES

Student_ID	Student_Name	Student_Age	Student_Gender
1	ONE	19	M
2	TWO	23	M

Here, each row is an entity.

Example :

1 one 19 M

RELATIONSHIP

1. Any association between two entity types is called a relationship.
2. Entities take part in the relationship. It is represented by a diamond shape.
3. For example, A teacher teaches students. Here, "teaches" is a relationship and this is the relationship between a Teacher entity and a Student entity.

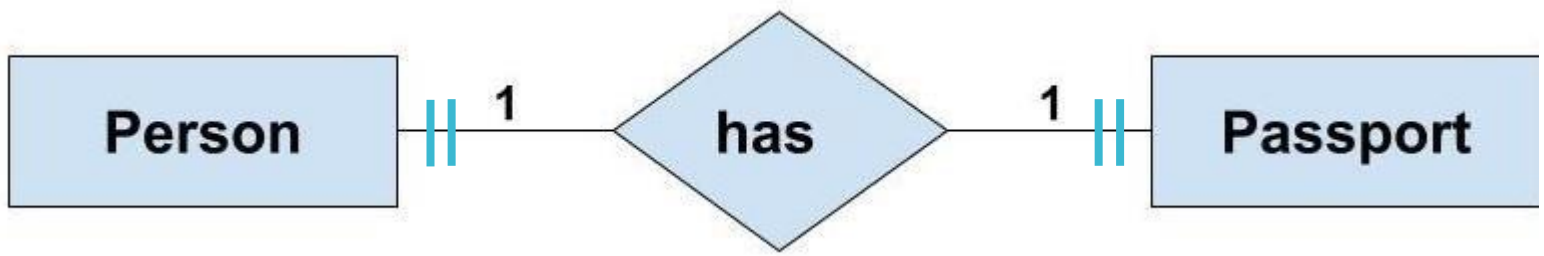
Three types of relationships can exist between two entities.

1. One-to-One Relationship
2. One-to-Many or Many-to-One Relationship
3. Many-to-Many Relationship

RELATIONSHIP

1. One-to-One Relationship

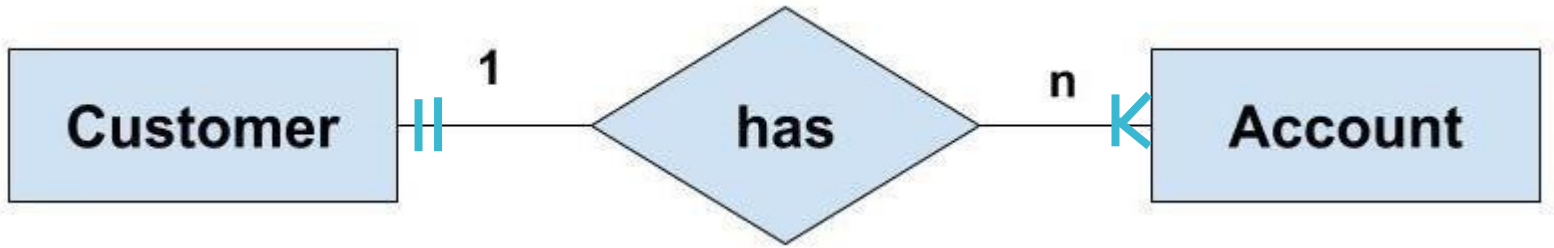
If there are two entities 'Person' (Id, Name, Age, Address)and 'Passport'(Passport_id, Passport_no). So, each person can have only one passport and each passport belongs to only one person.



RELATIONSHIP

2. One-to-Many Relationship

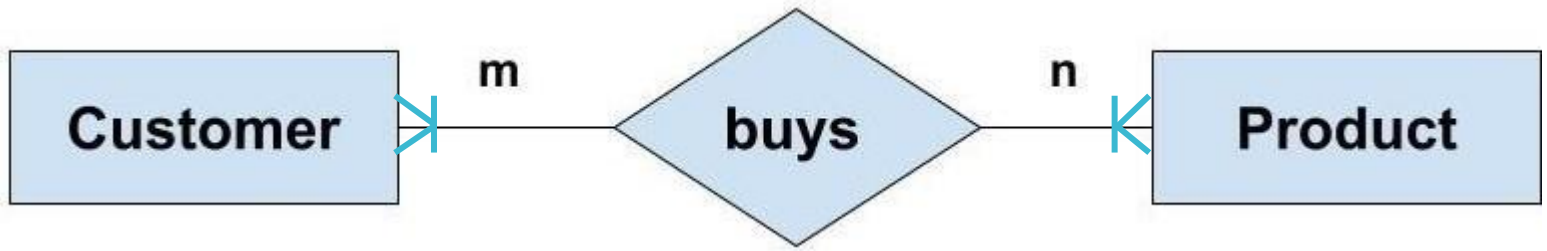
For example, If there are two entity types 'Customer' and 'Account' each 'Customer' can have more than one 'Account' but each 'Account' is held by only one 'Customer'.



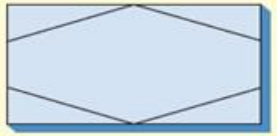
RELATIONSHIP

3. Many-to-Many Relationship

If there are two entity types 'Customer' and 'Product' then each customer can buy more than one product and a product can be bought by many different customers.



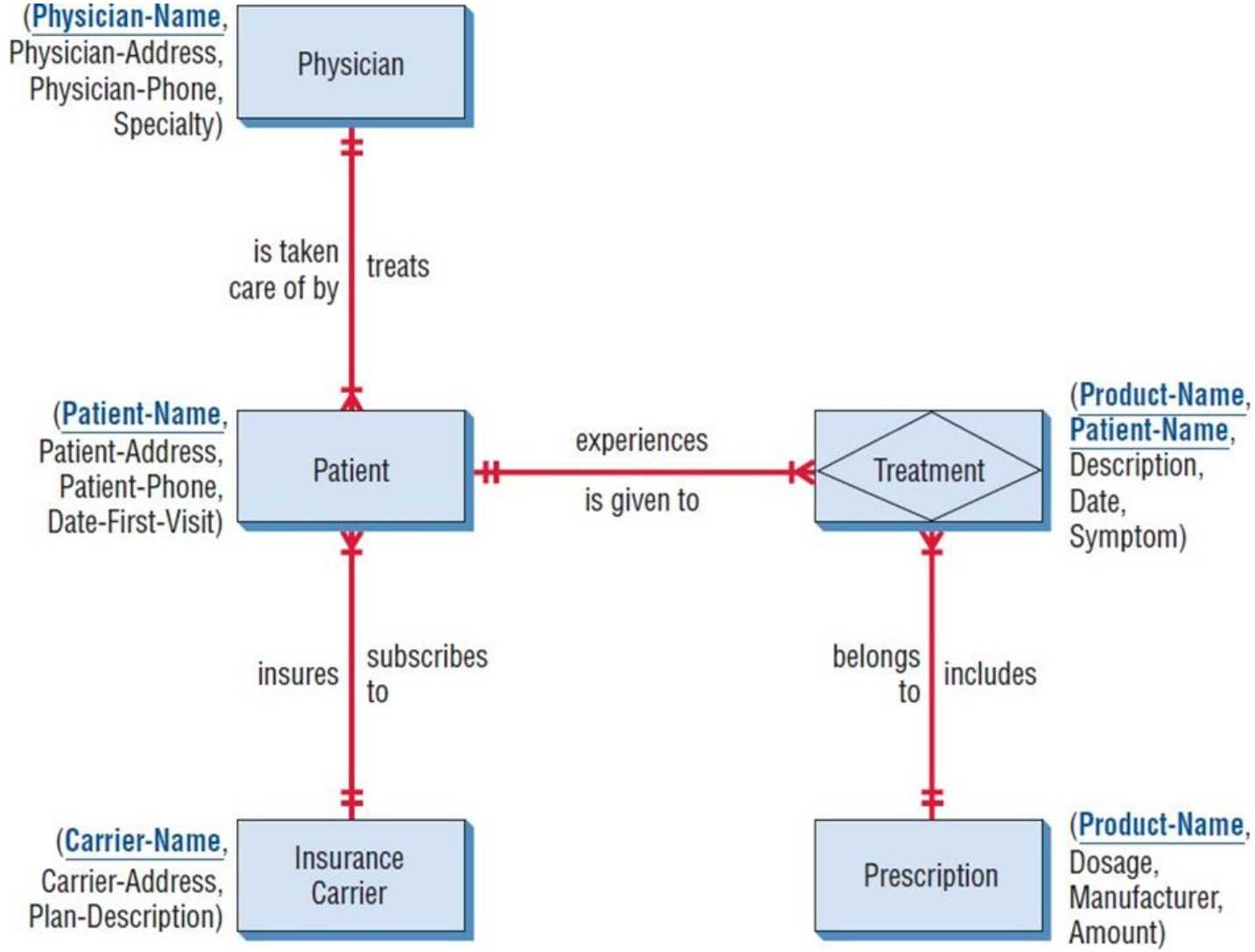
RELATIONSHIP (ER DIAGRAM SYMBOLS)

Symbol	Official Explanation	What It Really Means
	Entity	A class of persons, places, or things
	Associative entity	Used to join two entities
	Attributive entity	Used for repeating groups
	To 1 relationship	Exactly one
	To many relationship	One or more
	To 0 or 1 relationship	Only zero or one
	To 0 or more relationship	Can be zero, one, or more
	To more than 1 relationship	Greater than one

RELATIONSHIP ER DIAGRAM EXAMPLE

Here, a PHYSICIAN treats many PATIENT(s) (1:M), who each subscribe to an individual INSURANCE CARRIER. Of course, the PATIENT is only one of many patients that subscribe to that particular INSURANCE CARRIER (M:1). Many PATIENT(s) experience many TREATMENT(s), making it a many-to-many (M:N) relationship. TREATMENT(s) can include the taking of PRESCRIPTION(s), and thus is also an M:N relationship.

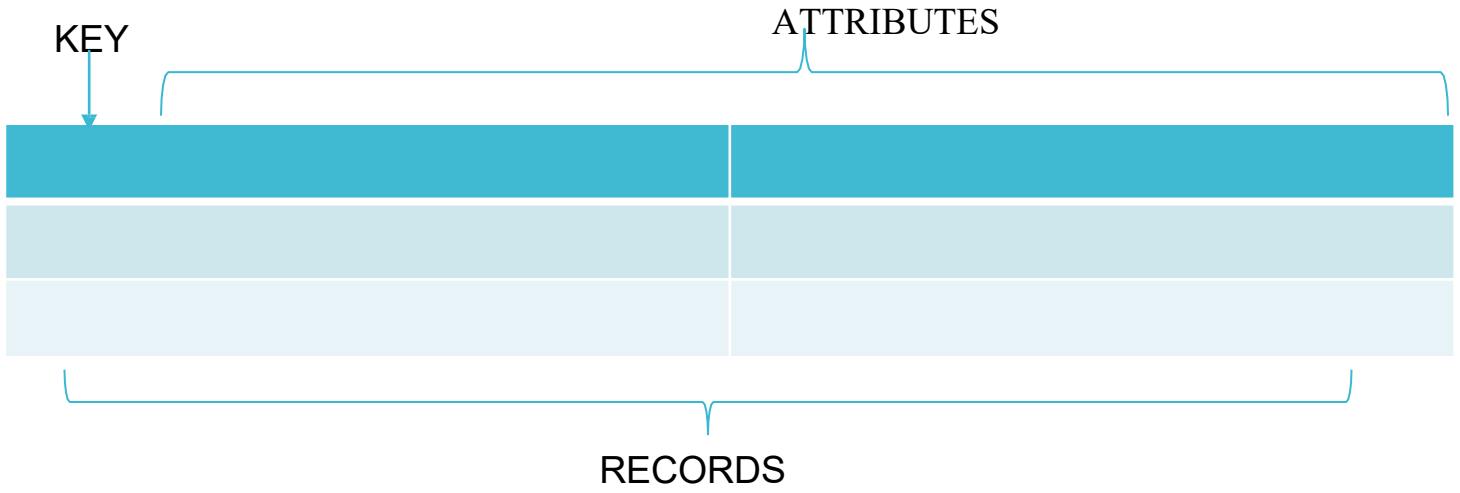
RELATIONSHIP ER DIAGRAM EXAMPLE



ATTRIBUTES, RECORDS AND KEYS

An attribute is some **characteristic of an entity.** There can be many attributes for each entity.

For example, a patient (entity) can have many attributes, such as last name, first name, street address, city, state, and so on.. To uniquely identify the student, we use the primary key as a roll number as it is not repeated.



RELATIONAL DATA STRUCTURE

ITEM PRICE

ITEM-#	TITLE	PRICE
B235	Guys and Dolls	8.99
B521	My Fair Lady	6.99
B894	42nd Street	10.99
B992	A Chorus Line	10.99

ORDER

ORDER-#	LAST NAME	I	STREET ADDRESS	CITY	ST	CHARGE ACCT
10784	MacRae	G	2314 Curly Circle	Lincoln	NE	45-4654-76
10796	Jones	S	34 Dream Lane	Oklahoma City	OK	44-9876-74
11821	Preston	R	1008 Madison Ave.	River City	IA	34-7642-64
11845	Channing	C	454 Harmonia St.	New York	NY	34-0876-87
11872	Kiley	R	765 Dulcinea Drive	La Mancha	CA	65-8798-87

ITEM STATUS

ITEM-#	ORDER-#	STATUS
B235	10784	Shipped 5/12
B235	19796	Shipped 5/14
B235	11872	In Process
B521	11821	In Process
B894	11845	Backordered
B894	11872	Shipped 5/12
B992	10784	Shipped 5/12

- A relational data structure consists of one or more two dimensional tables, which are referred to as relations. The rows of the table represent the records, and the columns contain attributes.

NORMALIZATION

A large database defined as a single relation may result in data duplication. This repetition of data may result in:

- Making relations very large.
- It isn't easy to maintain and update data as it would involve searching many records in relation.
- Wastage and poor utilization of disk space and resources.
- The likelihood of errors and inconsistencies increases. So to handle these problems, the NORMALIZATION PROCESS IS USED.
- Normalization is the process of organizing the data in the database. Normalization is used to minimize the redundancy from a relation or set of relations. It is also used to eliminate undesirable characteristics like Insertion, Update, and Deletion Anomalies. The normal form is used to reduce redundancy from the database table. Normalization consists of a series of guidelines that helps to guide you in creating a good database structure.

NORMALIZATION(1NF)

1 NF:

- There are only Single Valued Attributes.
- Attribute Domain does not change.
- There is a unique name for every Attribute/Column.
- The order in which data is stored does not matter.

Course	Content
Programming	Java
Programming	c++
Web	HTML
Web	PHP
Web	ASP

EXAMPLE –

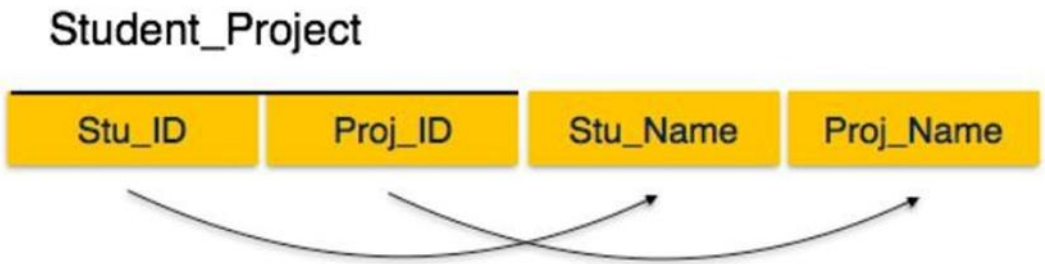
We re-arrange the relation (table) as below, to convert it to First Normal Form.

Course	Content
Programming	Java, c++
Web	HTML, PHP, ASP

NORMALIZATION(2NF)

2 NF:

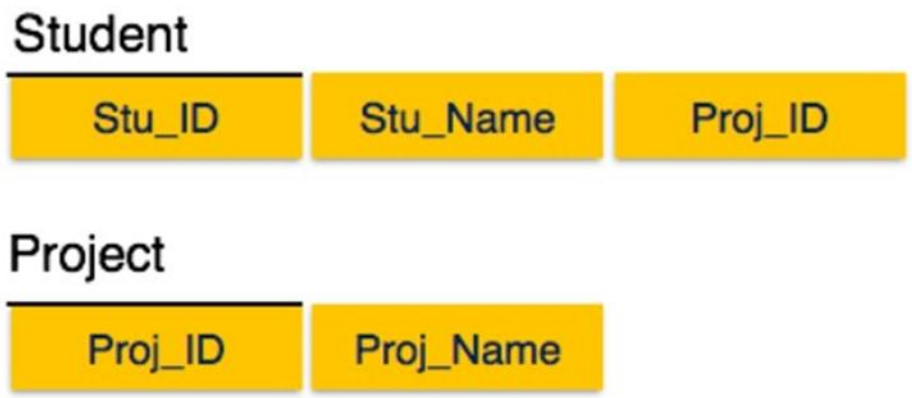
- In the second normal form, all the attributes will be functionally dependent on the primary key. Therefore, the next step is to remove all the partially dependent attributes and place them in another relation.



NORMALIZATION(2NF)

EXAMPLE –

In Student_Project relation that the prime key attributes are Stu_ID and Proj_ID. According to the rule, non-key attributes, i.e. Stu_Name and Proj_Name must be dependent upon both and not on any of the prime key attribute individually. But we find that Stu_Name can be identified by Stu_ID and Proj_Name can be identified by Proj_ID independently. This is called partial dependency, which is not allowed in Second Normal Form.



NORMALIZATION(3NF)

3 NF:

- A normalized relation is in the third normal form if all the non key attributes are fully functionally dependent on the primary key and there are no **transitive** (non key) dependencies.



NORMALIZATION(3NF)

EXAMPLE –

- We find that in the above Student_detail relation, Stu_ID is the key and only prime key attribute. We find that City can be identified by Stu_ID as well as Zip itself. Neither Zip is a superkey nor is City a prime attribute. Additionally, Stu_ID → Zip → City, so there exists transitive dependency.



ANOMALIES

- There are three types of anomalies that occur when the database is not normalized. These are – Data Redundancy, Insertion, update and deletion anomaly. Let's take an example to understand this.

- Example: Suppose a manufacturing company stores the employee details in a table named employee that has four attributes: emp_id for storing employee's id, emp_name for storing employee's name, emp_address for storing employee's address and emp_dept for storing the department details in which the employee works. At some point of time the table looks like this: The below table is not normalized. We will see the problems that we face when a table is not

ANOMALIES

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

ANOMALIES (DATA REDUNDANCY)

Data Redundancy Anomalies:

- Data redundancy occurs when the same data are stored in more than one place in the database (except for primary keys stored as foreign keys). This problem is solved by creating tables that are in 3NF.

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

ANOMALIES (Insertion)

INSERTION Anomalies:

- Suppose a new employee joins the company, who is under training and currently not assigned to any department then we would not be able to insert the data into the table if emp_dept field doesn't allow nulls.

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

ANOMALIES (updation)

UPDATION Anomalies:

In the above table we have two rows for employee Rick as he belongs to two departments of the company. If we want to update the address of Rick then we have to update the same in two rows or the data will become inconsistent. If somehow, the correct address gets updated in one department but not in other then as per the database, Rick would be having two different addresses, which is not correct and would lead to inconsistent data.

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

ANOMALIES (deletion)

DELETION Anomalies:

- Suppose, if at a point of time the company closes the department D890 then deleting the rows that are having emp_dept as D890 would also delete the information of employee Maggie since she is assigned only to this department.

emp_id	emp_name	emp_address	emp_dept
101	Rick	Delhi	D001
101	Rick	Delhi	D002
123	Maggie	Agra	D890
166	Glenn	Chennai	D900
166	Glenn	Chennai	D004

DE - NORMALIZATION

- It is an optimization strategy that is used after normalization has been achieved. Let's understand it with an example. Imagine a fruit seller has a daily list of what fruit is in stock in their stand and a daily list of the market prices for all fruits and vegetables. This would be two separate tables in a normalized database. If a customer wanted to know the price of an item, the seller would need to check both lists to determine if it is in stock and at what price. This would be slow and annoying.

- Therefore, every morning, the seller creates another list with just the items in stock and the daily price, combining the two lists as a quick reference to use throughout the day. This would be a denormalized table for speeding up reading the data.

DE - NORMALIZATION

Fruit_ID	Fruit_name	In stock(KG)	(KG)	Price(KG)
1	Apple	20	10	200
2	Mango	30	20	600

Veg_ID	Veg_name	In stock(KG)	(KG)	Price(KG)
1	Tomato	20	10	90
2	Potato	30	20	30

DE - NORMALIZATION

- An important consideration for normalizing data is if the data will be read heavy or write heavy. Because data is duplicated in a denormalized database, when data needs to be added or modified, several tables will need to be changed. This results in slower write operations.

Denormalization pros

- Faster reads for denormalized data, Simpler queries for application developers, Less compute on read operations.

Denormalization cons:

- Slower write operations, Additional database, Additional storage required for redundant tables

DE - NORMALIZATION

Veg/fruit name	In stock(KG)	(KG)	Price(KG)
Tomato	20	10	90
Potato	30	20	30
Apple	20	10	200
Mango	30	20	600

GUIDELINES FOR MASTER FILE/DATABASE RELATION DESIGN

1. The Each separate data entity should create a master database table. Do not combine two distinct entities on one file. For example, items are purchased from vendors. The ITEM MASTER table should contain only item information, and the VENDOR MASTER table should contain only vendor information.

2. Specific data field should exist only on one master table. For example, the CUSTOMER NAME should exist only on the CUSTOMER MASTER table, not on the ORDER table or any other master table. The exceptions to this guideline are the key or index fields, which may be on as many tables as necessary. If a report or screen needs information from many tables, the indexes should provide the linkage for obtaining the required records.

GUIDELINES FOR MASTER FILE/DATABASE RELATION DESIGN

3. Each master table or database relation should have programs to Create, Read, Update, and Delete (abbreviated CRUD) the records. Ideally, only one program should add new records and only one program should delete specified records. Many programs, however, may be responsible for changing data fields in the course of normal business activities. For example, a CUSTOMER MASTER file may have a CURRENT BALANCE field that is increased by the ORDER TOTAL in the order processing program and decreased by a PAYMENTAMOUNT or an AMOUNT RETURNED from two additional programs.

DATA WAREHOUSES

- Data warehouses differ from traditional databases. A Data Warehouse (DW) is a relational database that is designed for query and analysis rather than transaction processing. It includes historical data.

- The purpose of a data warehouse is to organize information for quick and effective queries. It stores a huge amount of data, which is typically collected from multiple sources like files, DBMS, etc. **The goal is to produce statistical results that may help in decision makings and to see quick different result.** For example, **Social Media Websites :** The social networking websites like Facebook, Twitter, Linkedin, etc. are based on analyzing large data sets. These sites gather data related to members, groups, locations, etc., and store it in a single central repository. Being a large amount of data, Data Warehouse is needed for implementing the same.

DIFFERENCE BETWEEN DATA WAREHOUSES AND TREDITIONAL DATA BASE

- Database Is designed to record Data and DW purpose is designed to analyze Data stored in the Database is up to date. And in DW, Current and Historical Data is stored . It May not be up to date.
- Simple transaction queries are used in DATABASE and in DW, Complex queries are used for analysis purpose.
- In a data warehouse, data are organized around major subjects rather than individual transactions.
- Data in a data warehouse cover a much longer time frame than data in traditional transaction-oriented databases because queries usually concern longer-term decision making rather than daily transaction details.

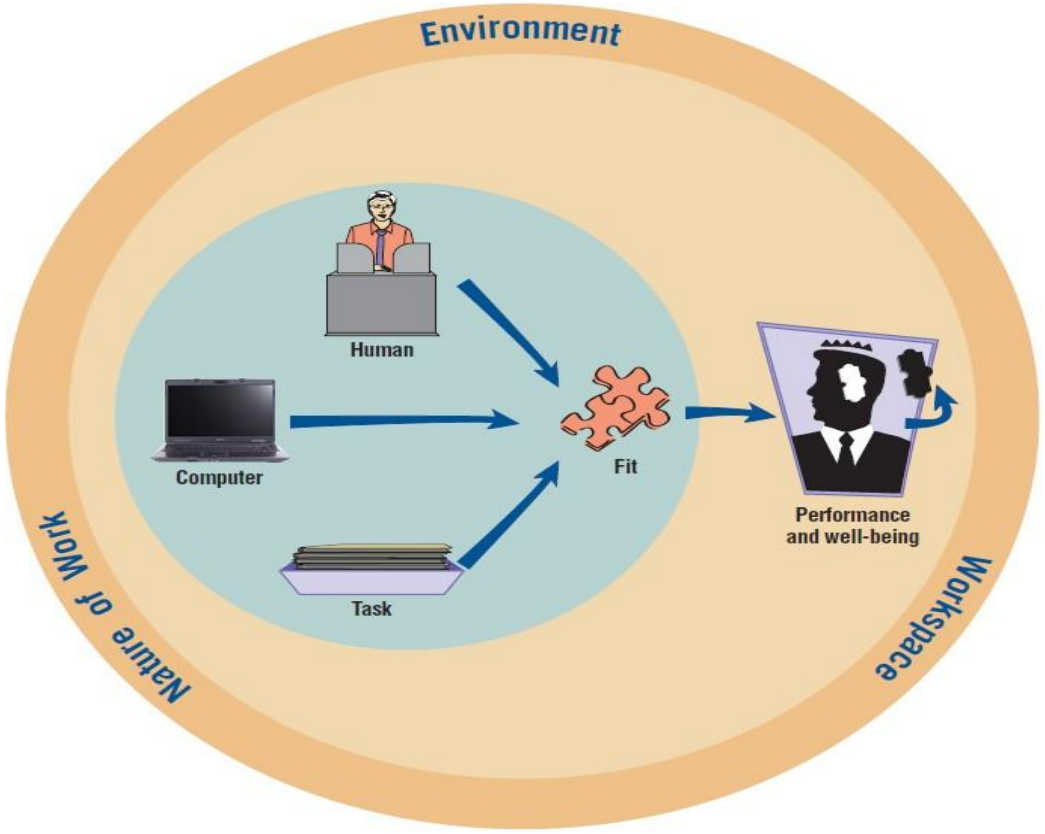
DIFFERENCE BETWEEN DATA WAREHOUSES AND TREDITIONAL DATA BASE

- Most data warehouses are organized for fast queries, whereas the more traditional databases are normalized and structured in such a way as to provide efficient storage of information.
- Data warehouses are usually optimized for answering complex queries, known as OLAP, from managers and analysts, rather than simple, repeatedly asked queries.
- Data warehouses include not just one but multiple databases that have been processed so that the warehouse's data are defined uniformly. These databases are referred to as clean data.

HCI (HUMAN COMPUTER INTERACTION)

- HCI means "Ensuring system functionality and usability, providing effective user interaction support, and enhancing a pleasant user experience." Furthermore, "The overarching goal is to achieve both organizational and individual user effectiveness and efficiency. To reach these goals, managers and developers need to be knowledgeable about the interplay among users, tasks, task contexts, information technology (IT), and the environments in which systems are used".

HCI (HUMAN COMPUTER INTERACTION)



HCI (HUMAN COMPUTER INTERACTION)

- The "fit" between the HCI elements of the human, the computer, and the **task** that needs to be performed leads to **performance** and **well- being.** Better fit is meant to result in better performance and greater overall well-being for the human involved in the system. Fortunately, humans' capacity to learn better ways to work also influences the fit. Users can be trained to develop a better fit by learning their tasks and computers thoroughly. Training continues to be an important way to improve fit.

HCI (HUMAN COMPUTER INTERACTION)

- **Task :** One has to learn many methods that help you understand, document, and graphically depict the tasks that people currently perform in the organization. We have also learned methods to help us design new tasks that will help people reach their objectives with the new systems you are creating. Tasks can be structured and routine, or they can be ill defined and without apparent structure. Complex tasks that require human, system, and task interaction are supported by ecommerce and Web systems, ERP systems, and wireless systems inside and outside the organization.

HCI (HUMAN COMPUTER INTERACTION)

- **PERFORMANCE :** The definition of the word performance in the HCI context is also key. In this case, the term performance refers to a combination of the efficiency involved in performing a task and the quality of the work that is produced by the task. For example, if analysts are using high level software or a CASE tool to create data flow diagrams in which they are proficient, we would predict that the quality of the data flow diagrams produced would be high. The performance is also efficient, because the analysts are using an automated tool with which they are familiar. They can work rapidly, with good results.

HCI (HUMAN COMPUTER INTERACTION)

- **WELL-BEING :** At this point, we can introduce the concept of well-being, which is a concern for a human's overall comfort, safety, and health; in sum, it is their physical as well as psychological state.

HCI (HUMAN COMPUTER INTERACTION)

- Does using a CASE tool for producing UML diagrams or DFDs on a computer serve the analyst's well-being? Yes, because the task fits well with the analyst, the software, the objective, and the computer. Notice that the analysts are working in an environment where they are physically comfortable, are psychologically stimulated to be creative, and can be productive; also, each analyst's work is valued by peers and clients, as well as valued monetarily by the employing organization.

THE TECHNOLOGY ACCEPTANCE MODEL(TAM) AND ATTITUDE

- TAM is a way for analysts to organize their thinking about whether users will accept and use information technology.

- Some researchers add an explicit attitude dimension to their idea of the technology acceptance model that helps them think more specifically about what psychological states will shape the way users accept or reject the use of the information systems they design.

- Most of the HCI research looks at a variety of user attitudes, including satisfaction, anxiety, enjoyment, and playfulness in approaching technology.

USABILITY

- Usability studies (according to www.useit.com) are all about finding out what works in the world and what doesn't. The ISO has created usability standards that you can explore on www.usabilitynet.org/tools/r_international.htm.

- The standards cover the use of the product (effectiveness, efficiency, and satisfaction in a particular context of use), the user interface and interaction, the process used to develop the product, and the capability of an organization to apply user-centered design.

USABILITY

- Nielsen and Mack (1994) and Nielsen, Molich, Snyder, and Farrell (2001) have published usability heuristics (or rules of thumb) based on thousands of usability tests of interfaces and, later, tests of ecommerce Web sites. They include visibility of system status, match between the system and the real world, user control and freedom, consistency and standards, error prevention, reconnection rather than recall, flexibility and efficiency of use, aesthetic and minimalist design, help that users recognize, diagnosis and recovery from errors, and help and documentation

DESIGNING FOR THE COGNITIVE STYLES OF INDIVIDUAL USERS (USABILITY)

- One important consideration is that data, particularly data used for decision making, are made available in different forms so that users with different cognitive abilities can make sense of them.

- Some users may prefer to examine tables and make decisions, some prefer graphs, and others want to read narrative text.

DESIGNING FOR THE COGNITIVE STYLES OF INDIVIDUAL USERS (USABILITY)

- **Making sure data is made available in different forms**

1. Tables (A data table (in excel) is a straight up Listing with columns and rows. It could be useful for simple reports.)
2. Pivot tables (A pivot table include different type of aggregation like average, sum, count etc. We can apply additional filters use to perform data analysis)
3. Graphs
4. Text
5. Different times

PHYSICAL CONSIDERATIONS IN HCI DESIGN (USABILITY)

While creating website designing, we learned the basis for sound design of screens, forms, Web sites, and databases. This included the special use of fonts, color, and layout design to communicate to users and to help them do the right thing with the input and output. For that, it includes 3 considerations :

1. **Vision** Factors such as length of the distance from display to the person performing a task; the angle of the display in relation to the person viewing it; the size and uniformity of the characters; the brightness, contrast, balance, and glare of the screen; and whether a display is blinking or stable can all be designed to standards established through ISO and other national and international groups.

PHYSICAL CONSIDERATIONS IN HCI DESIGN (USABILITY)

2. **Hearing** Noisy laser printers, phone conversations, and shredders can lead to overload on human hearing. Office workers can wear noise-canceling headphones or get a personal music player like an iPod. As an analyst you will need to consider noise when you design office systems.

3. **Touch** When using an HCI perspective to evaluate the usefulness of keyboards and other input devices

CONSIDERING HUMAN LIMITATIONS, DISABILITIES, AND DESIGN (USABILITY)

All humans have limitations in their physical capabilities. Some are immediately visible, others are not. When designing from an HCI perspective, you start realizing that limitations are often discussed in terms of disabilities.

As a systems analyst you will be subject to the legal provisions of the country in which you are working. For example, if you are designing for workplaces in the United States, you may want to access the obligations of an employer under the Americans with Disabilities Act at www.eeoc.gov/types/ada.html.

CONSIDERING HUMAN LIMITATIONS, DISABILITIES, AND DESIGN (USABILITY)

An employer in the United States is expected to make reasonable accommodation to employ a disabled person, which includes "Making existing facilities used by employees readily accessible to and usable by persons with disabilities; job restructuring, modifying work schedules, reassignment to a vacant position; acquiring or modifying equipment or devices, adjusting or modifying examinations, training materials, or policies, and providing qualified readers or interpreters."

IMPLEMENTING GOOD HCI PRACTICES (GUIDELINES)

Guidelines for the HCI Approach to Systems Design

- Examine the task to be done and consider the fit among the human, computer, and task.
- Identify what obstacles exist for users in their attempts to accomplish their assigned tasks.
- Keep in mind the perceived usefulness and perceived ease of use from TAM.
- Consider usability. Examine the usage environment by creating use case scenarios that depict what is going on between users and the technology.
- Use the information you have gained beforehand to figure out the physical and organizational environmental characteristics. Design with prototyping to accommodate diverse users and users with disabilities.

INTERFACE DESIGN (OBJECTIVES)

- It is important to recognize that the interface is the system for most users.

- Your goal must be to design interfaces that help users and businesses get the information they need in and out of the system by addressing the following **objectives:**

1. Matching the user interface to the task.
2. Making the user interface efficient.
3. Providing appropriate feedback to users.
4. Generating usable queries.
5. Improving the productivity of computer users.

INTERFACE DESIGN (TYPES OF INTERFACE)

- Natural-language interfaces
- Question-and-answer interfaces
- Menus
- Form-fill interfaces
- Command-language interfaces
- Graphical User Interfaces (GUIs)

GUIDELINES FOR DIALOG DESIGN

There are several key points for designing good dialog. They include the following:

1. Meaningful communication, so that the computer understands what people are entering and people understand what the computer is presenting or requesting.
2. Minimal user action.
3. Standard operation and consistency.

MEANINGFUL COMMUNICATION

- The system should present information clearly to the user. This means having an appropriate title for each display and providing clear user feedback. For Example - Displaying slashes between the month, day, and year in a date field or commas and decimal points in an amount field. In a graphical user interface, the cursor may change shape depending on the work being performed. Remember to use icons or text in addition to color coding in order to reach the largest number of users. Many GUIs incorporate tool tip help, displaying a small help message identifying the function of a command button when the cursor is placed over it.

FEEDBACK FOR USERS

- All systems require feedback to monitor and change behavior.
- Feedback compares current behavior with predetermined goals and gives back information describing the gap between actual and intended performance.
- Feedback also increases human confidence.
- As designers of user interfaces, systems analysts need to be aware of the human need for feedback and build it into the system
- Web sites should display a status message or some other way of notifying the user that the site is responding and that input is either correct or in need of further information.

FEEDBACK FOR USERS [TYPES OF FEEDBACK]

- Acknowledging acceptance of input (Example - A Web example would be a Web page displaying a message that "Your payment has been processed")

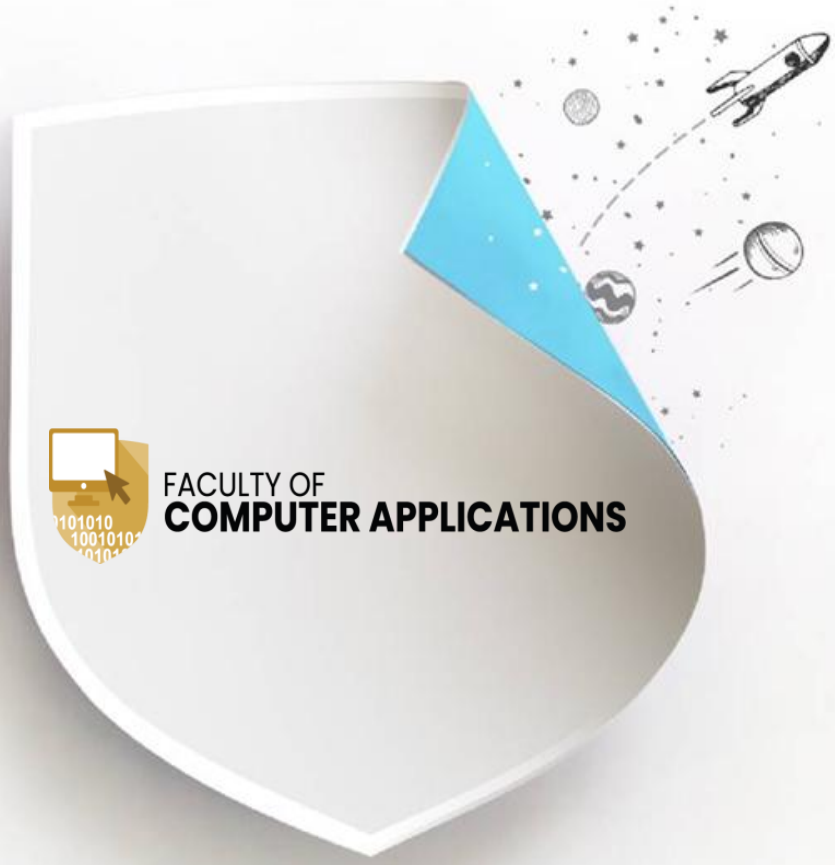
- Recognizing that input is in the correct form (For example, a user inputs a command, and the feedback states "READY" as the program progresses to a new point.)

- Notifying that input is not in the correct form (For example – Please enter proper value)

FEEDBACK FOR USERS [TYPES OF FEEDBACK]

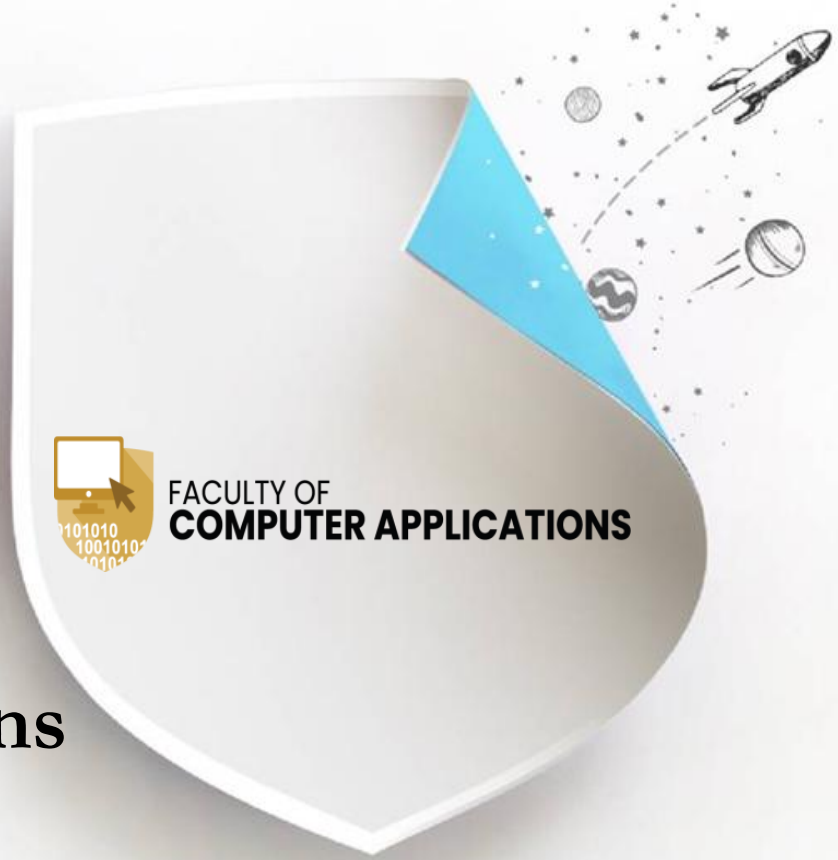
- Explaining a delay in processing (Example – wait/Loading msg)
- Acknowledging that a request is completed (Example - Employee record has been added)
- Notifying that a request was not completed (Example - "Unable to process request. Check request again,")
- Offering the user more detailed feedback(Example - the user may type a question mark or click on an appropriate icon to get more feedback. Using the command Help as a way to obtain further information has been questioned)

THANK YOU



Marwadi University
Marwadi Chandarana Group

NAAC A+



FACULTY OF COMPUTER APPLICATIONS

Bachelor of Computer Applications

BCA Semester: 2

05BC3202 – Analysis and Design of Systems